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Improving Diabetes Care and Health Measures Among Hispanics Using Community Health Workers: Results From a Randomized Controlled Trial

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The increasing prevalence of diabetes and obesity, growing health disparities, and shortage of bilingual and culturally trained health care professionals underscore the role of trained community health workers (CHWs) to provide economically sustainable and culturally relevant services. This prospective randomized design evaluated the relative effectiveness of a CHW intervention among Hispanic persons with newly diagnosed type 2 diabetes, as compared with usual clinic practice in three inner-city health centers. In sum, 189 Hispanic patients newly diagnosed with type 2 diabetes were randomly assigned to one of three 6-month diabetes management approaches—CHW, case management, and standard provider care—and assessed for diabetes-related health measures and clinical indicators at baseline and postintervention. Participants in the CHW group achieved greater improvements than did the controls in program measures: health status, emergency department utilization, dietary habits, physical activity, and medication adherence. They also had 2.9 times greater odds of decreasing body mass index.

Keywords: *health disparities; diabetes; randomized trial*

The importance of diabetes control rests in the prevention of adverse diabetes-related complications that contribute to early mortality, morbidity, reduced quality of life, and excess health care costs (Zhang, Engelgau, Norris, Gregg, & Narayan, 2004).

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The prevalence of diabetes in the United States has been increasing since the middle of the last century, and it continues to accelerate (Centers for Disease Control and Prevention, 2007). Studies indicate that the prevalence of obesity, a contributing factor to diabetes, has also risen significantly over the past 20 years (Mokdad, Ford, Bowman, Dietz, Vinicor, & Bales, 2003). Given the growing epidemics of diabetes and obesity, particularly among certain ethnic populations, there is impetus to define and implement culturally sensitive interventions aimed at helping to improve health status and the quality of diabetes care.

Diabetes control is an elusive goal in most populations, but it is even more challenging in the Hispanic community (Davidson, 2003). Hispanics have the highest lifetime risk of developing diabetes (45.4% in males and 52.5% in females) when compared to non-Hispanic Whites and non-Hispanic Blacks (Narayan, Boyle, Thompson, Sorensen, & Williamson, 2003). Various studies on the health of Hispanic populations in the United States have documented significant disparities in diabetes prevalence, treatment outcomes (McBean, Huang, Virnig, Lurie, & Musgrave, 2003), access to health services, including receipt and timeliness of care, as well as disparities in the clinical encounter. Furthermore, obesity among Hispanics continues to increase, placing these individuals at risk for diabetes and diabetes-related complications.

Different interventions have been reported to improve the care of underserved populations, including Hispanics with diabetes. These approaches include patient-level interventions (such as education and self-management programs), provider-level interventions (including treatment algorithms), disease and care/case management programs, health system changes (Glazier, Bajcar, Kennie, & Wilson, 2006), and the use of community health workers (CHWs; Brown, Garcia, Kouzekanani, & Hanis, 2002).

CHWs—also known as lay health educators, community health advocates, community health outreach workers, community health representatives, and *promotores de salud*—are members of the community who provide services to help connect the health care system with the residents living within the area. The inherent goal of using CHWs in underserved communities is to form an alliance between health care providers and patients to enhance access, appropriate utilization of health care services, and reduce health risks. CHWs have been used in countries around the world, including the United States, for decades in the areas of HIV education, immunization programs, high-risk maternal outreach initiatives, and cancer-screening campaigns. Over the past several years, literature supporting CHWs in diabetes management has been reported; however, many of these reports were observational, addressed specific interventions (such as education only), and more recently, integrated CHWs into diabetes programs (Brown et al., 2002; Philis-Tsimikas, Walker, Rivard, Talavera, Reimann, & Salmon, 2004).

This study advances current knowledge of diabetes management by evaluating the relative effectiveness of an intervention delivered by CHWs on various health measures and clinical indicators among Hispanic persons newly diagnosed with type 2 diabetes, as compared to usual clinic diabetes care.

METHOD

Design

A prospective randomized pre–post measurement design was used to assign patients newly diagnosed with type 2 diabetes to one of three study groups: CHW, case

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management, or standard provider care. To assess potential contamination of the standard provider care group and to minimize the possibility that program impact could be attributable to the Hawthorne effect, clinical indicators were abstracted for a noncontact control group of type 2 diabetes patients who were not enrolled in the study and who received care in clinics that did not participate in the study. The noncontact control group was selected from patient records post hoc, using inclusion criteria. This group comprised the clinical records of 137 patients who met the study's eligibility requirements but were never recruited for participation, because they did not receive care at participating clinics. These patients did not include those who declined participation. The research protocol was approved by an external institutional review board. The program was completed in January 2004.

Participants

Participants were recruited during routine clinic visits at three inner-city family health centers in Los Angeles, between July 2002 and July 2003. Inclusion criteria were as follows: Hispanic/Latino by self-report, 18 years of age or older, and diagnosis of type 2 diabetes (via American Diabetes Association [ADA] criteria) within 6 months of study enrollment. Patients with gestational diabetes or previous diabetes case management were excluded.

During the recruitment period, 1,352 patients with diabetes mellitus received care at the clinics. Of those, 354 patients were eligible to participate on the basis of inclusion criteria (race/ethnicity, age, and diagnosis criteria). Of those eligible, 31 (9%) did not complete the baseline survey within the enrollment period, and 5 (1%) refused to participate by not signing a consent form. The final study sample consisted of 318 patients randomly assigned to the three study arms via a random-number table.

Description of Diabetes Management Programs

All participants received standard provider care by licensed physicians and/or nurse practitioners. Standard provider care included routine clinic appointments, laboratory tests, medications, and referrals as necessary. In addition, a standardized packet of diabetes education materials, presented in both Spanish and English and tailored for the local Hispanic population, was provided during the initial visit during the study period regardless of group assignment.

CHW. The primary goal of the CHW program, Amigos en Salud (Friends in Health), was to implement a culturally acceptable intervention composed of diabetes education and monitoring services that incorporated individual participant preferences using trained laypeople.

The study intervention was delivered by three full-time bilingual Hispanic CHWs who had diabetes or had experienced it through a family member or friend. The CHW staffing-to-patient ratio over the 6-month period was 1 to 35 patients. They were recruited from the surrounding community, clinic, and local organizations. Because the CHWs were paid clinical staff, they were required to meet organizational requirements for education, which required a high school degree or a GED degree. Two of the CHWs were educated in their country of origin, either Mexico or Central America.

All CHWs received formal program training before the study was initiated. The 6-week training curriculum included clinic polices and procedures, CHW roles and

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responsibilities, diabetes standards, self-management strategies incorporating patient cultural and spiritual beliefs, and health behavior change theory. Over the 6-month intervention period, CHWs conducted individual educational sessions based on ADA standards with participants and their family members. The 10-week ADA education sessions were tailored to the participants' needs, such as knowledge, identified problems, goals, and level of progress. Thus, some topics may have required reinforcement and multiple sessions, whereas other topics may not have been necessary.

CHWs routinely made follow-up telephone calls to participants to monitor self-management progress, identify barriers and issues, and assist in problem solving. The CHW intervention strategy was based on the transtheoretical (stages of change) model (Prochaska & Velicer, 1997). CHWs used culturally appropriate educational materials based on stages of change to help participants improve their diabetes self-management skills. CHWs assessed patient understanding, addressed knowledge deficits, and evaluated health behaviors, such as frequency and intensity of physical activity, dietary habits, medication adherence, and self-monitoring practices. Education, positive reinforcement, and tips on achieving health behavior goals were provided by CHWs using the program education materials. Education topics were based on a standardized curriculum, and sessions were conducted in accessible community locations, at the clinic, or in the patient's home. The mean number of CHW-conducted sessions (in person or via telephone contact) was 11.3 during the 6-month intervention period.

Case Management. Patients in the case management group received diabetes care and education in the clinic setting from two linguistically competent and culturally sensitive registered nurses, as an adjunct to standard provider care. Each nurse care manager managed an average of 53 study patients—in addition to their other patients with chronic diseases, such as asthma—who were referred to them as part of the clinic's case management program. Case managers worked directly with individual patients, following standardized clinic protocols for diabetes education and monitoring based on ADA clinical recommendations. Patients under case management were usually seen on a monthly basis in the clinic or as needed. Follow-up calls were made, as determined by the case manager. Other responsibilities included patient assessment, development of a case management treatment plan incorporating provider treatment, coordination and referral of community resources, and participation in multidisciplinary case conferences to discuss patient status.

Standard Provider Care. Patients assigned to the standard provider care group received only standardized clinical care by physicians and nurse practitioners, without case management or CHW services.

MEASURES

Survey Instruments

Participants in the three randomized groups completed an identical behavioral questionnaire at baseline and at 6-month program completion. Behavioral data were collected via a standardized survey instrument completed by the patient or administered by the CHW or case management assistant, as needed. Survey questions were drawn from existing validated instruments, including the 2002 Behavioral Risk Factor Surveillance

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System, from the Centers for Disease Control and Prevention; the Morisky Self-Reported Medication Behavior Scale, to assess medication-taking behavior and adherence (Morisky, Ang, Krousel-Wood, & Ward, 2008); and the Diabetes Knowledge Questionnaire, a measure of diabetes knowledge that is available in both English and Spanish and that is used to help facilitate the CHW education sessions (Garcia, Villagomez, Brown, Kouzekanani, & Hanis, 2001).

Clinical Endpoints

Clinical measures assessed at baseline and follow-up included A1c and body mass index (BMI) measurements. Baseline measures were required within 3 months before study enrollment, and follow-up measurements were taken no less than 6 months after enrollment.

Data Analysis

Patients who completed both baseline and 6-month follow-up questionnaires were included in the analysis. Descriptive statistics and analysis of variance compared demographics, influential characteristics, clinical indicators, and self-reported health measures between groups at baseline and 6-month follow-up. Paired *t* tests investigated within-group changes over the follow-up period. Multivariate logistic regression models evaluated the effect of study group and other covariates on changes in A1c and BMI at follow-up. Mediational analyses were conducted to investigate whether physical activity and diet mediated the relationship between study group and decrease in BMI at follow-up. Mediation is where a predictor has an indirect effect on an outcome through a third variable. This indirect effect may account for some (or all) of the main effect of the predictor on the outcome. Mediational analyses were conducted according to the principles outlined by Baron and Kenny (1986). According to which, association between the independent variable, the dependent variable, and the potential mediator is a prerequisite for proceeding with meditational analysis. Logistic regression analyses were used to investigate these associations. All logistic regression analyses controlled for gender and age because they are important predictors of BMI decrease. If the study group, the hypothesized mediating variable (diet or exercise), and a decrease in BMI were all associated with each other, then we proceeded to investigate the extent to which the hypothesized mediating variable mediated the relationship between study group and decrease in BMI, by observing changes in the magnitude of the study group effect on decrease in BMI when the variable was added to the model. If the magnitude of the study group effect on decrease in BMI was attenuated after the hypothesized mediating variable was added to the model, a mediating relationship was determined to exist.

RESULTS

A total of 189 patients (59%) successfully completed the program; 129 patients moved out of the area, disenrolled from the study, or could not be contacted during the follow-up period. Loss to follow-up was greatest in the standard provider care group. Significantly greater proportions of patients enrolled in the standard provider care group (50%) and the case management group (43%) were lost to follow-up, as compared with the CHW group (28%, $p < .05$). When compared to males, a greater proportion of

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eligible females (63%) completed the program (50%, $p < .05$). With respect to age and annual income, no significant differences were observed between program graduates and those who disenrolled. Demographics of the study participants were also comparable to those of the patient population at the three clinics during the time of the study.

Sample characteristics were consistent with the demographics of Hispanic patients at the clinics where participants were recruited (Table 1). Of the 189 patients who completed the program, the majority were females (64%); the mean age was 50 years (range = 18-87 years); and most reported having less than a sixth-grade education (61%). Of the 104 patients (55%) who provided household income data, 101 (97%) reported an annual household income of less than $25,000. There were no significant differences across study groups with respect to age, education, and household income; however, when compared with the other groups, the standard provider care group had a greater proportion of females, and the case management group had a greater proportion of patients whose parents had diabetes.

Baseline clinical indicators and self-reported health measures indicated a population characterized by poor dietary and exercise habits, as well as poor medication-taking behavior, with no significant differences at baseline across study groups (Table 2). The average period between study enrollment and baseline measurements for A1c and BMI was 0.13 and 0.04 months, respectively. The average period between baseline and follow-up measurements for A1c and BMI was 9.6 and 7.7 months, respectively. All groups were similar with respect to the periods between each clinical measurement.

Whereas no between-group differences were observed at baseline, at 6-month follow-up significant between-group differences were observed in health status, emergency department (ED) admissions, medication-taking behavior, fruit and vegetable intake, exercise, and diabetes knowledge (Table 2). At follow-up, the proportion of patients in the CHW group who reported "very good" or "excellent" health increased from 5% to 57% ($p < .05$). In contrast, no significant changes in health status were observed in the case management or standard provider care group. Self-reported ED admissions in the previous 6 months did not change significantly in the CHW and case management groups. By contrast, ED admissions increased significantly from 13% to 28% ($p < .05$) in the standard provider care group. Medication-taking behavior was assessed in patients who were receiving therapy, and it remained unchanged in the CHW group. By contrast, medication-taking behavior worsened in the case management and standard provider care groups. The proportion of patients who reported never forgetting to take their medication decreased significantly in both the case management group (77% to 55%, $p < .05$) and standard provider care groups (67% to 50%, $p < .05$). The proportion of patients who reported consuming two or more servings of fruits and vegetables daily increased significantly in the CHW and case management groups but not in the standard provider care group. Patients' self-reported intake of fatty foods decreased significantly from 29% to 16% ($p < .05$) in the CHW group but remained unchanged in the other groups. Physical activity improved significantly across all groups except the case management group. Patients who reported exercising at least 3 days a week increased from 28% to 63% ($p < .05$) in the CHW group and from 17% to 35% ($p < .05$) in the standard provider care group but remained unchanged in the case management group.

Mean A1c decreased from 8.6% to 7.2% ($p < .05$) in the CHW group, 8.5% to 7.4% ($p < .05$) in the case management group, and 9.5% to 7.4% ($p < .05$) in the standard provider care group. In contrast, decreases in A1c values were not observed among the noncontact control group (7.5% to 7.7%, $p > .05$).

None of the group-level changes in average BMI was statistically significant. Mean BMI at baseline and follow-up was 32.5 and 32.0 ($p > .05$) in the CHW group, 32.2 and

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Table 1. Demographics and Other Characteristics at Baseline by Study Group

Variables	Total (N = 189)	Community Health Worker (n = 75)	Case Management (n = 60)	Standard Provider Care (n = 54)	p
Age					.82
M (*SD*)	50.0 (11.9)	51.0 (12.5)	50.0 (12.1)	50.0 (11.0)	
Mdn (Range)	51 (18-87)	53 (18-87)	49 (21-83)	51 (26-74)	
	n (%)	*n* (%)	*n* (%)	*n* (%)	
Female	121 (64)	48 (64)	31 (52)	42 (78)	.01
Completed less than sixth grade	116 (61)	50 (67)	35 (58)	31 (57)	.48
Household income					
Less than $25,000	101 (53)	41 (55)	30 (50)	30 (56)	.96
$25,000 or more	3 (2)	1 (1)	1 (2)	1 (2)	
Preferring not to answer	85 (45)	33 (44)	29 (48)	23 (43)	
Primary language is Spanish	181 (96)	73 (97)	59 (98)	49 (91)	.20
Married/living together	98 (52)	39 (52)	34 (57)	25 (46)	.88
Mother or father has diabetes	84 (45)	34 (45)	32 (55)	18 (35)	.04
Diagnosed with high cholesterol	87 (47)	33 (45)	25 (43)	29 (54)	.70

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Table 2. Changes in Clinical Indicators and Self-Reported Health Measures by Study Group

	Community Health Worker (n = 75)		Case Management (n = 60)		Standard Provider Care (n = 54)	
Clinical Indicator	Baseline n (%)	6-Month Follow-Up n (%)	Baseline n (%)	6-Month Follow-Up n (%)	Baseline n (%)	6-Month Follow-Up n (%)
Good health in the previous 4 weeks[a]	4 (5)	43 (57)*	5 (8)	12 (20)	4 (7)	12 (22)
Admitted to emergency department in the previous 6 months†	15 (20)	7 (9)	9 (15)	12 (20)	7 (13)	15 (28)*
Never forgetting to take their diabetes medication†	52 (69)	59 (79)	46 (77)	33 (55)*	36 (67)	27 (50)*
Two or more servings of fresh fruit per day†	35 (47)	55 (73)*	30 (50)	47 (78)*	28 (52)	29 (54)
Two or more servings of fresh vegetables per day†	29 (39)	57 (76)*	19 (32)	34 (57)*	19 (35)	23 (43)
Two or more servings of fatty foods per day	22 (29)	12 (16)*	15 (25)	14 (23)	22 (41)	17 (32)
Exercise at least three times per week†	21 (28)	47 (63)*	15 (25)	23 (38)	9 (17)	19 (35)*
Mean Diabetes Knowledge Scale score†	10.6	14.7*	10.6	11.0	10.3	10.6
Mean A1c (%)[a]	8.6	7.2*	8.5	7.4*	9.5	7.4*
Mean body mass index (kg/m^2)[b]	32.5	32.0	32.2	32.5	31.2	31.5

a. Community health worker group, n = 56; case management group, n = 48; standard provider care group, n = 43.
b. Community health worker group, n = 58; case management group, n = 37; standard provider care group, n = 31.
†$p < .05$ between groups at 6-month follow-up. *$p < .05$ within groups at 6-month follow-up.

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32.5 ($p > .05$) in the case management group, and 31.2 and 31.5 ($p > .05$) in the standard provider care group. However, when compared to those of the standard provider care group, patients in the CHW group had 2.9 times greater odds (95% confidence interval [CI] = 1.3-6.2) of decreasing their BMI (Table 3). This association persisted after controlling for gender, age, and dietary and exercise habits. Age was also associated with decreased BMI. Specifically, after controlling for study group, gender, and dietary and exercise habits, patients 50 years and older were less likely (odds ratio [OR] = 0.4, 95% CI = 0.2-0.8) to have reduced BMI at follow-up. We observed a positive association between exercise frequency and decreased BMI in the univariate model (OR = 2.2, 95% CI = 1.1-4.1); however, this association did not persist after the other covariates were controlled for.

Neither diet nor exercise met all the criteria for mediation between study group and decrease in BMI (Figure 1). Study group was associated with both diet and exercise at 6-month follow-up. Compared with the standard provider care group, patients in the CHW group were more likely (OR = 2.43; 95% CI = 1.13-5.23) to report having two or more servings of fresh fruit per day. Also, compared with the standard provider care group, patients in the CHW group were more likely (OR = 2.87, 95% CI = 1.34-6.17) to report exercising three or more times per week. However, neither diet (OR = 1.94, 95% CI = 0.94-4.00) nor exercise (OR = 1.89, 95% CI = 0.98-3.63) was associated with decrease in BMI; thus, both failed to meet all the criteria for mediation.

DISCUSSION

These findings suggest that a culturally tailored outreach and education program, based on the transtheoretical model and delivered by trained CHWs, can significantly improve self-care behaviors and decrease BMI, when compared to case management and standard provider care. Many participants in this study had limited financial and nonfinancial resources. Studies show that under such circumstances, individuals may delay or avoid seeking medical care, filling prescriptions, or adhering to treatment regimens because they must deal with more pressing day-to-day issues (Weinick, Jacobs, Stone, Ortega, & Burstin, 2004). Health becomes a low priority amid their daily challenges. In spite of these issues, the CHW program was effective at improving health status and behaviors.

The association between overweight/obesity, physical inactivity, dietary habits, and the development of type 2 diabetes is well documented (Sullivan, Morrato, Ghushchyan, Wyatt, & Hill, 2005). Improving these behaviors can lead to improved health benefits among patients with diabetes (Brown et al., 2002; Sullivan et al., 2005). The CHW intervention had the greatest impact on these modifiable risk factors, as compared with the other intervention groups. In addition, greater improvements in medication-taking behavior and reductions in ED utilization were observed in the CHW group. Studies have demonstrated that patients with poor adherence to medication regimens are at increased risk of hospitalizations (Morisky et al., 2008) and diabetes-related complications (Yong-Fang et al., 2003). Furthermore, empowering patients to better manage their diabetes and appropriately reduce ED utilization is important among Hispanic patients, who typically experience barriers to care and suffer disproportionately from diabetes complications (Weinick et al., 2004). The greater reductions in self-reported ED visits among the CHW group may be attributed to the effectiveness of the CHWs in improving participants' ability to navigate the health care system, proactively access health care, improve self-management skills, and seek preventive care.

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Table 3. Univariate and Multivariate Associations Between Study Characteristics and Decrease in Body Mass Index Among Groups

		Univariate		Multivariate	
Study Characteristic	%	OR	95% CI	OR	95% CI
Study group					
Standard provider care	28.5	1.0		1.0	
Community health worker	39.6	2.9*	1.3-6.2	2.6*	1.1-6.0
Case management	31.7	0.9	0.4-2.2	0.8	0.3-2.0
Number of servings of fresh fruit (exit)					
One or less per day	29.6	1.0		1.0	
Two or more per day	70.4	1.8	0.9-3.7	1.9	0.9-4.3
Exercise frequency (exit)					
Two times or fewer per week	50.0	1.0		1.0	
Three times or more per week	50.0	2.2*	1.1-4.1	1.4	0.7-2.8
Age					
<50 years old	44.0	1.0		1.0	
≥50 years old	56.0	0.4*	0.3-0.8	0.4*	0.2-0.8
Gender					
Male	36.0	1.0		1.0	
Female	64.0	0.9	0.5-1.8	0.8	0.4-1.8

NOTE: OR = odds ratio; CI = confidence interval. Odds ratios calculated from multiple logistic regressions for the association with each characteristic within the population, adjusted for the other characteristics in the table; the reference category is the first row within each characteristic.
*Denotes a statistically significant result at the $p < .05$ level (or better).

Prior studies have demonstrated the efficacy of self-management programs delivered by trained lay personnel among other patient populations, in other settings, with other chronic conditions (Fedder, Chang, Curry, & Nichols, 2003; Philis-Tsimikas et al., 2004). However, this study, conducted in the Los Angeles metropolitan area, is the first randomized controlled trial comparing a CHW intervention for Hispanics with diabetes to case management and usual provider care. The results provide insight into the impact of a CHW intervention on Hispanics with diabetes in an inner-city community. They also fulfill some of the current limitations in research outlined by Brownstein and colleagues (2005), by extending the model into underserved high-risk populations and formal assessment of CHW performance. Modest reductions in BMI have been shown to improve glycemic control and insulin sensitivity, including reductions in A1c, fasting plasma glucose, and cardiovascular risk (Anderson, Kendall, & Jenkins, 2003). Our results therefore suggest important benefits of participation in the CHW program regarding diabetes management. It is noteworthy that we report a significant effect of exposure to the CHW program on decrease in BMI, above and beyond the effects of exercise and diet. This finding suggests that exposure to the CHW program resulted in improvements in some mediators that we did not measure or include in our analyses. Although we did not find enough evidence to support even partial mediation of exercise and diet on the relationship between study group and decrease in BMI, such a casual pathway makes intuitive sense. It is probable that our

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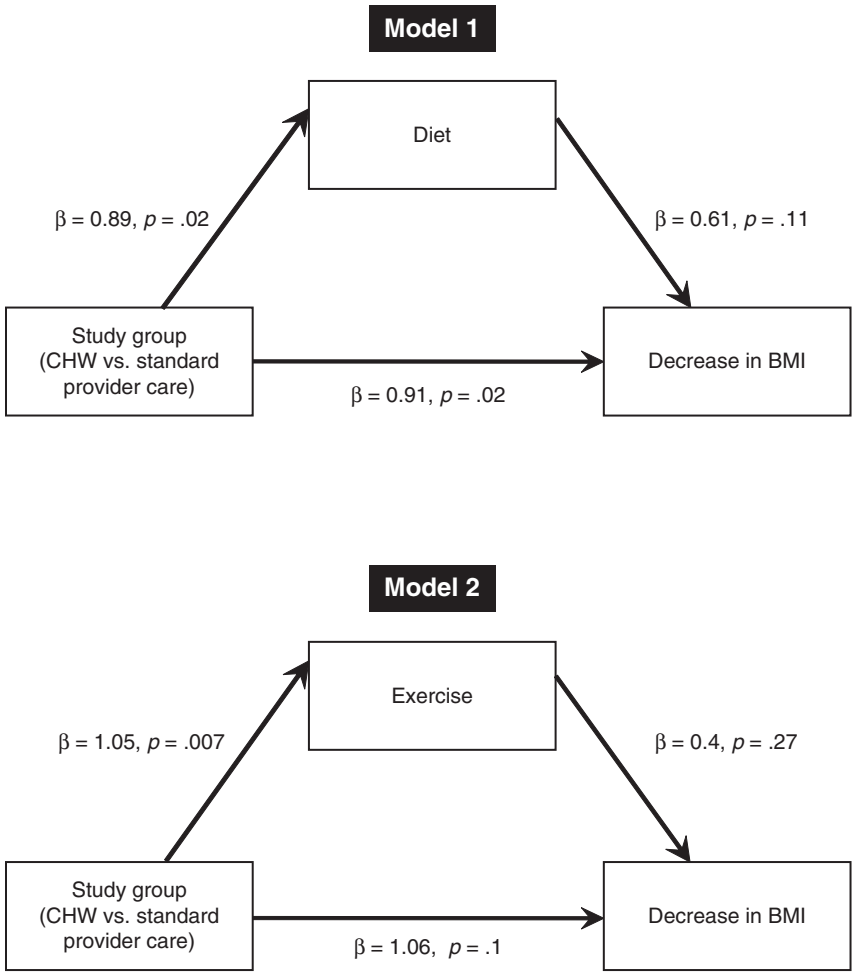

Figure 1. Diet and exercise as potential mediators of the effect of study group on decrease in body mass index.
NOTE: CHW = community health worker.

small sample size resulted in insufficient power to detect the meditational effect of diet and exercise on BMI within our hypothesized causal pathway.

In addition to improvements in behavioral and clinical results, the CHW intervention group had a higher proportion of patients completing the study. As a result of the high frequency of contact between CHWs and patients, CHWs become trusted and accessible members of the health care system who can support follow-up and retention—namely, through their role as patient advocates who connect patients and family members to the health care system. Similar findings have been reported and attributed to the use of CHWs through increased social support and as visible members of the health care team (Corkery et al., 1997).

Other interventions utilizing a CHW model have reported reductions in hospital and ED expenditures, although the results were not statistically significant (Fedder et al.,

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2003). Decreased utilization, as noted in this study and elsewhere, contributes to the potential relative cost-effectiveness of this intervention. Further financial evaluations are necessary to demonstrate the overall impact of the CHW model.

Limitations of this study include the use of self-reported measures to assess health behaviors and ED utilization. In addition, the short follow-up period may not have been sufficient to fully evaluate sustained self-care behaviors and clinical indicators (A1c, BMI). The standard provider care group, as well as the other randomized groups, may have received higher levels of care during the study period than what they would have normally received in the absence of the intervention. Before study activities began, clinic providers received information about the study objectives. Furthermore, each eligible participant required provider approval to enroll in the study. Only in the case of noncontact controls were providers blinded to the study's objectives and activities. It is conceivable that awareness of the study among clinic providers may have motivated them to be more diligent about the services they offered to study patients. This improved provider care might explain the decrease in A1c observed across all three randomized groups, in contrast to the increase experienced by the noncontact controls. Complete medication history was not available for all participants, which may have had direct impact on the differences in baseline A1c—specifically, the lower baseline A1c in the noncontact control group. The short intervention period and the program's relatively small sample size may have limited statistical power to detect some intervention effects, which may explain, for example, why the observed decrease in mean BMI from 32.5 to 32.0 within the CHW group was not statistically significant.

Our study findings further highlight the value of using bicultural/bilingual CHWs, providing formal CHW training in diabetes practice standards and health behavior, and utilizing educational materials tailored to the patient's readiness to change and literacy level. The CHW model has been supported by the American Association of Diabetes Educators (2003), American Public Health Association (2002), and the Centers for Disease Control and Prevention (2003).

This study provides empirical support that an outreach model utilizing trained CHWs and culturally and literacy-appropriate educational materials can have a significant effect on patient behaviors and health status outcomes. The results suggest that employing CHWs in the delivery of the diabetes intervention can fortify usual clinic practices (case management and standard provider care), which arguably require more resources because they rely on highly trained personnel. CHWs cannot replace health care professionals; however, they can augment ongoing clinic practices, particularly in settings where resources are limited and staffing shortages exist. The linguistic competency of the CHWs and the incorporation of individualized cultural beliefs and behaviors into the CHW intervention compose a critical value-added component. These factors are important in understanding the mechanism of social and behavioral determinants that lead to diabetes control among ethnically diverse populations.

CHW programs have been implemented in a variety of culturally diverse communities to support management of chronic conditions and increase access to health care. With the increased prevalence of diabetes and obesity, rising health care costs, and shortage of bilingual and culturally trained health care professionals, the use of trained CHWs should be considered as health policies evolve that better align incentives for providing economically sustainable and culturally relevant health worker services. Future research into the economic impact of the CHW model should be investigated to assess program sustainability and inform health policy decisions.

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IMPLICATIONS FOR PRACTICE

CHWs function to help reach underserved populations who have limited access to health care services by connecting patients, family members, and the community. Through self-management education and training, social support, and community engagement, CHWs can help to address and reduce the impact of diabetes and its complications within the traditional health care delivery system. As members residing within the community, CHWs have a greater understanding and knowledge of the local cultural values, such as health beliefs and behaviors, access issues to health services, social and environmental influences, and language barriers. Besides having an understanding of community life, CHWs can provide additional assistance in retention and follow-up, helping patients to appropriately engage and utilize the health care system and local community resources, thus resulting in informed, activated patients and family members. With these attributes, they can support health care professionals in changing health behaviors and improving the management of diabetes. In return, health care professionals can support CHWs by establishing specific roles and responsibilities for CHWs within clinics, ensuring adequate CHW training, and continuing to provide medical management of diabetes patients.

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